STUDENT VERSION: Group 12-Month Macroeconomic Strategy Project

Project Objective: Deliver a 12-month U.S. Macroeconomic Forecast for an Investment Committee (IC). You will use institutional data (FRED/FactSet), manual trend analysis, and AI "Red Teaming" to build a defensible outlook.

Save the Excel File with a Team Name. Share the document among the team

Fill out the Team Information Tab in the given Excel Workbook.

Including selecting a role and responsibility for each student.

Role	Primary Responsibility	Key Deliverable
1. Data Architect	Excel Modeling. Manages the Macro_Starter_Template.xlsx.	Initial Analysis Sensitivity Matrix
2. Consensus Analyst	FactSet Research. Gathers the "Street" view via FactSet.	FactSet Consensus Range & Thesis Justification Table.
3. AI Orchestrator	Prompt Engineering. Manages all AI "Red Teaming."	AI Audit Log & narrated Bear Case timeline.
4. Strategy Lead	Synthesis & IC Memo. Finalizes the investment thesis.	Final Investment Committee Memo & Provide a "Bottom Line Up Front" (BLUF)

Phase 1: Human vs. AI "Regime" Analysis

Objective: Before asking the AI for answers, your team must understand the 5-year baseline.Complete the Phase 1 Worksheet in the Excel Workbook

Step 1.1: The Initial Assessment (Team Task)

Open the `Historical Data` tab. As a group, identify and record the following trends from the last 5 years. In the Excel Workbook -complete the `Phase 1 Worksheet` `(questions are also listed below)`

Phase 1 Worksheet: Initial Estimate

Group Name: _____

Data Architect: _____

Instructions: Open the Historical Data tab in the Macro Starter File. As a group, review the last 5 years of data. Do not use AI for this worksheet. Your goal is to identify the "momentum" of the economy to provide a baseline for your 12-month forecast.

Section 1: The Growth & Inflation Pulse

Analyze the relationship between Real GDP and CPI.

1. ***The Growth Peak: Identify the quarter with the highest Real GDP growth. What was the rate? _____%***
2. ***The Inflation Peak: Identify the month where CPI (Inflation) was at its highest. What was the rate? _____%***
3. ***The Trend Observation: Looking at the last 4 quarters, is GDP accelerating, decelerating, or stagnating?***
   Observation:
   _____

Section 2: The Leading Indicators

Analyze the Yield Curve and Consumer Sentiment.

1. ***Yield Curve Inversion Check: Scan the 10Y-2Y Spread column.***
   - ***Is the spread currently Positive (Normal) or Negative (Inverted)?***
     _____
   - ***How many months in the last 2 years has the curve been inverted?***
     _____

2. ***Sentiment vs. GDP: Find a period where Consumer Sentiment dropped significantly. Did Real GDP drop in the same quarter, or did GDP stay high while people felt worse?***
   Observation:
   _____

Section 3: The Monetary Environment

Analyze the Federal Funds Rate ($FEDFUNDS$).

1. ***The Hiking Cycle: When did the most aggressive increase in interest rates begin? (Date: _____)***
2. ***The Plateau: Has the Fed Funds Rate stayed at the same level for the last 6 months, or is it still moving?***
   Observation:
   _____

Section 4: Preliminary Regime Hypothesis

Based ONLY on your manual audit above, circle the "Regime" you believe describes the current economy.

- ***Expansion: High Growth, Low/Stable Inflation, Positive Yield Curve.***
- ***Late-Cycle: Decelerating Growth, High Inflation, Flat/Inverted Yield Curve.***
- ***Recession: Negative Growth, Falling Inflation, High Unemployment.***
- ***Recovery: Low but Rising Growth, Falling Inflation, Steepening Yield Curve.***

Team Justification:

"We believe the regime is _____ because..."

_____
_____
_____

Step 1.2: The AI Regime Validation (AI Orchestrator)

Upload the file and use this "Chain-of-Thought" prompt:

> ***"I have uploaded 5 years of U.S. macro data. Perform the following: 1) Identify the current 'Economic Regime' (e.g., Late-Cycle, Recovery, Stagflationary). 2) Compare the current Fed Funds Rate to the 10Y-2Y Spread; what does historical precedent suggest about growth 12 months from now? 3) Identify any 'Divergence' where sentiment and GDP are moving in opposite directions."***

Students should place their Preliminary Regime Hypothesis (side-by-side with the AI's identified regime.

Variable	*Human View (Worksheet)*	*AI View (Validation)*	*Final Group Consensus*
Current Regime	*e.g., Expansion*	*e.g., Late-Cycle*	*Late-Cycle*
Primary Risk	*Inflation*	*Yield Curve Inversion*	*Yield Curve Inversion*
Sentiment	*"Confusingly low"*	*"Reflecting high rent"*	*High rent/cost of living*

- ***The Match: If both say "Late-Cycle," the team should look for the AI's reasoning. Does the AI cite the Yield Curve while the students cited Inflation?***
- ***The Mismatch: If the students said "Expansion" but the AI said "Late-Cycle," the AI Orchestrator must ask: "We identified growth as accelerating, but you labeled this Late-Cycle. Which specific leading indicator are you seeing that we missed?"***

Phase 2: FactSet Navigation & Thesis Justification

Task: The Consensus Analyst must locate the "Street Consensus" for the 12-month outlook.

How to find data in FactSet:

1. **Consensus Estimates: Type `@ECO` in the top search bar.**
    - **Set Country to "United States."**
    - **Select the Estimates tab.**
    - **Look for Real GDP and CPI. Note the Mean, High, and Low for the next 4 quarters.**
2. **Economic Data Explorer: Type `@ECON` to find deep historical trends for Consumer Sentiment or Treasury Spreads if you need to supplement the FRED data.**
3. **Yield Curve: Type `@YIELD` to see a live visual of the 10Y-2Y spread.**

4. Create a Team forecast based on your Initial Economic estimates and the AI feedback. Add to the Numerical Expectations tab

Variable	Current Level (FRED)	Market Consensus (FactSet/CME)	Team Forecast (12-Mo)	Thesis Direction Vs Market Consensus	Justification & AI Red-Team Defense
Real GDP Growth	e.g., 2.1%	@ECO Mean: _____	_____%	[Above / Below / Neutral]	Identify the "Primary Driver." Summarize how you defended this number against the AI's "Red Team" critique.

Fed Funds Rate	**e.g., 5.33%**	**CME FedWatch: _____**	**_____%**	**[Hawkish / Dovish / Neutral]**	**Why do you believe the Fed will cut more or less than what the market currently prices in?**
Consumer Sentiment	**e.g., 68.0**	**@ECON Trend: _____**	**_____**	**[Optimistic / Pessimistic]**	**Explain the relationship between your Inflation forecast and this Sentiment number.**

Phase 3: The Sensitivity Matrix (Excel)

Step 3—Building the Matrix

In the real world, an Investment Committee (IC) doesn't just want to know what you think will happen; they want to know what happens to their money if you are wrong.

The matrix serves three critical functions:

1. **Stress Testing: It identifies the "breaking point" of the economy. At what specific interest rate level does growth turn negative?**
2. **Visualizing Correlation: It shows how two variables (Rates and Sentiment) can "gang up" on the economy.**
3. **Eliminating Bias: By forcing students to look at a 6x6 grid of outcomes, they can no longer ignore the "Bear Case" simply because they want to be "Bullish."**

Task: The Data Architect builds the "Risk Matrix" in the template.

4. **The Assumptions:**
    a. **Cell *B2* = *The Teams Effective Federal Funds Target Rate- Base Case***
    b. ***Cell B3* = *Your Teams Base Case Sentiment level***
5. **The Engine (Cell B5): Create the GDP formula using decimals:**
   `=0.02 + ((B3 - 65) * 0.001) - ((B2 - 0.0525) * 0.5)`

6. **The Skeleton: * Anchor: In D10, type = and click B5.**
    a. **Side Axis: List Rates in D11:D16 (e.g., 6.0% down to 4.5%).**
    b. **Top Axis: List Sentiment in E10:I10 (e.g., 55 up to 75).**
7. **The Tool: Highlight D10:I16. Go to Data > What-If Analysis > Data Table.**
    a. **Row Input Cell: Select B3 (Sentiment).**
    b. **Column Input Cell: Select B2 (Interest Rate).**
8. **Formatting: Apply Conditional Formatting > Color Scales (Red-Yellow-Green) to visualize the "Red Zone."**

   The matrix should look "something like" this …. with your calculations:

		Sentiment				
		55 (Low)	60 (Mid)	65 (Base)	70 (Strong)	75 (Bull)
Anchor	2.00%	55	60	65	70	75
Interest Rates	6.00%	0.63%	1.13%	1.63%	2.13%	2.63%
	5.75%	0.75%	1.25%	1.75%	2.25%	2.75%
	5.50%	0.88%	1.38%	1.88%	2.38%	2.88%
	5.25%	1.00%	1.50%	2.00%	2.50%	3.00%
	5.00%	1.13%	1.63%	2.13%	2.63%	3.13%

How to Build a Professional Bear Case using the Matrix

The "Bear Case" is not just a random bad outcome; it is a data-driven narrative derived from the "Red Zone" of your team's sensitivity matrix.

1. Locate the "Red Zone"

After running the Data Table tool and applying Conditional Formatting, you will see a cluster of red cells, usually in the top-left corner (where Interest Rates are highest and Sentiment is lowest).

2. Identify the "Trigger"

You should pick the most realistic "worst-case" cell in that Red Zone.

- *Example:* **If the Base Case is 4.5% Rates and 70 Sentiment, a realistic Bear Case "Trigger" might be 5.5% Rates and 55 Sentiment.**

3. Create the "AI-Orchestrated" Narrative

Instead of guessing why things went wrong, the AI Orchestrator uses the specific numbers from that red cell to force the AI to build a logical timeline.

The Bear Case Prompt: *"In our Excel model, a scenario of 5.5% Interest Rates combined with 55 Consumer Sentiment results in a GDP contraction of -0.4%. Act as a Bear Market Strategist. Write a 12-month 'Hard Landing' timeline explaining the Transmission Mechanism:

1. **How do 5.5% rates specifically trigger corporate defaults in the next 6 months?**
2. **How does the 55 Sentiment level lead to a 'Retrenchment' in consumer spending?**
3. **How do these two factors combine to reach that -0.4% GDP figure by Month 12?"***

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4. Create a summary table of your base case and bear case scenarios?

Feature	Base Case (Phase 2)	Bear Case (Phase 3)
Probability	50% - 60% (Most Likely)	20% - 25% (Tail Risk)
Excel Source	The Anchor Cell (B5)	The Red Zone of the Matrix
Market Driver	FactSet Consensus	A "Shock" to the Model
IC Focus	Where to invest?	Where to invest? Should we hedge for protection?

Phase 4: Formal Investment Committee Memo

The Investment Committee is composed of busy executives. They need the answer in the first 30 seconds. Create an IC memo showing all of the information in the next 4 steps (sample template provided below).

Step 1: Establish the "BLUF" (Bottom Line Up Front)

- **The Lead: State your 12-month outlook immediately. Are you "Bullish," "Bearish," or "Neutral"?**
- **The Numbers: Provide your specific 12-month targets for GDP Growth and CPI Inflation.**
- **The Logic: In one sentence, identify the "Primary Driver" (e.g., "The primary driver of our outlook is the contraction in bank credit availability despite stable consumer spending.")**

Step 2: Present the Thesis Justification Table

Insert the table completed in Phase 2. This proves your group didn't "hallucinate" numbers but grounded them in FactSet and FRED data.

- **The "Why": For each variable, the Strategy Lead must write 1-2 sentences explaining why the team chose a forecast that differs from the FactSet mean.**

Step 3: Conduct the "AI Audit" (Defense of the Base Case)

This is where the AI Orchestrator provides the documentation of the "Red Teaming" process.

- **The Critique: Describe the strongest argument the AI made *against* your group's thesis.**
- **The Rebuttal: Explain how the team responded. Did you change your numbers because the AI was right, or did you find FactSet data that proved the AI was wrong?**

Step 4: The "Red Zone" Narrated Bear Case

This is the most critical part of the risk analysis. It uses the Sensitivity Matrix from Phase 3 to define the "Downside Risk."

1. **Identify the Trigger: Select a specific "Red" cell from your Excel matrix (e.g., 6.0% Rates and 55 Sentiment).**
2. **Narrate the Timeline: Use the AI-generated timeline to explain the Transmission Mechanism.**
    - ***Month 1-4:* What event causes the shock?**
    - ***Month 5-8:* How do high rates "break" the consumer or the corporation?**
    - ***Month 9-12:* How do we arrive at the specific negative GDP number shown in that Excel cell?**

Step 5: Final Investment Recommendation

Translate the macro forecast into an investment strategy.

- **Asset Allocation: Based on your 12-month outlook, suggest a weighting for Equities vs. Bonds.**
- **Sector Selection: If you are "Bearish," suggest defensive sectors (e.g., Utilities, Staples). If you are "Bullish," suggest growth sectors (e.g., Tech, Industrials).**

<u>Investment Committee Template (this is the submission for this phase of the project)</u>

TO: Investment Committee (IC)

FROM: [Group Name / Analyst Names]

DATE: [Current Date]

RE: 12-Month U.S. Macroeconomic Outlook

1. Executive Summary (The Lead) Provide a "Bottom Line Up Front" (BLUF). State clearly if the economy will Expand, Contract, or Stagnate over the next 12 months. Summarize your target GDP and CPI numbers.

2. Thesis Justification [Insert the Table from Phase 2 here]. Explain why your forecast deviates from the FactSet mean.

3. Base Case Defense & AI Audit

- **The Argument: Explain the "Transmission Mechanism." How do current interest rates impact your 12-month GDP?**
- **The AI Audit: Document your "Red Teaming." What flaw did the AI find in your logic? Why did you choose to either accept or reject the AI's critique?**

4. Risk Analysis (Sensitivity Modeling) Discuss your Excel Matrix results. Identify the "Red Zone" (High Rates/Low Sentiment).

- **The Bear Case (25% Prob): Insert the 12-month "Hard Landing" timeline generated by the AI Orchestrator.**

5. Investment Recommendation Based on your macro outlook, should the IC be Overweight or Underweight Equities relative to Bonds?

Below is a "High Quality" example of the Investment Committee Memo. This reflects a high-pass submission based on the volatile macroeconomic environment of early 2022. This period was characterized by the initial spike in inflation, the start of the Fed's hiking cycle, and the geopolitical shock of the Ukraine conflict.

Example:

12-Month Investment Committee Memo

TO: Global Asset Allocation Committee

FROM: Macro Strategy Team (Group 4 with names)

DATE: March 15, 2022

RE: 12-Month U.S. Macroeconomic Outlook: "The Inflationary Inflection"

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1. Executive Summary (The Lead)

We project the U.S. economy will undergo a sharp deceleration (1.5% - 1.9% GDP growth) over the next 12 months. While 2021 was defined by a post-pandemic "boom," the 12-month outlook is clouded by headline CPI hitting 40-year highs. We recommend an Underweight position in growth equities and a shift toward Commodities and Defensive Value, as the "easy money" era of the Fed Funds Rate at 0% concludes.

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2. Thesis Justification Table

Our team's forecast is more "bearish" on growth and "hawkish" on inflation than the early 2022 FactSet consensus.

Variable	Current (FRED)	Market Consensus	Team Forecast	Thesis Direction	Justification & AI Defense

Real GDP	**6.9% (Q4 '21)**	**@ECO Mean: 3.5%**	**1.7%**	**Below**	**AI Red-Team argued that the "Great Reopening" would sustain 3%+ growth. We countered with FRED data showing a -2.5% drop in real wages, which will crush discretionary spending.**
Int. Rate	**0.08%**	**FedWatch: 1.50%**	**2.50%**	**Hawkish**	**Market consensus expects a "measured" pace. We believe the Fed is "behind the curve" and will be forced into 50bps hikes to combat 7.9% headline CPI.**

Sentiment	**59.4**	**@ECON Trend: 65.0**	**52.0**	**Pessimistic**	**Gas prices (approaching $4.50/gal) have a 0.85 negative correlation with sentiment. We expect a 12-month low in consumer confidence.**

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3. Base Case Defense & AI Audit

The Economic Logic: The primary driver of our 1.7% GDP forecast is the "Cost-Push Inflation" shock. The sudden spike in energy and fertilizer prices (exacerbated by the Ukraine conflict) acts as a functional tax on the U.S. consumer. With the 10Y-2Y Spread rapidly flattening (currently 0.25%), the margin of error for a "Soft Landing" has disappeared.

AI Audit Log (Sample Entry):

- **Initial Group View: We originally predicted 3.0% GDP, assuming inflation was "transitory."**
- **AI Red-Team Critique: *"You are ignoring the 'Wage-Price Spiral.' Historical data from the 1970s shows that once inflation enters the service sector, it becomes 'sticky.' Your growth forecast is too high because you aren't accounting for the Fed being forced to induce a slowdown to break inflation."***
- **Revision: We accepted the critique. We slashed our GDP forecast from 3.0% to 1.7% and shifted our Fed Funds assumption significantly higher.**

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4. Risk Analysis (Sensitivity Modeling)

Our Excel model indicates that the economy is extremely vulnerable to the "Double-Whammy" of high energy prices and rising borrowing costs. Our Base Case (2.50% Rates / 52 Sentiment) yields a 1.7% growth rate.

The Bear Case: The "Stagflationary Stall" Timeline (Prob: 25%)

Using the AI-generated timeline for our matrix's worst-case scenario (4.0% Rates / 45 Sentiment):

- **Months 1-4: Oil hits $130/barrel; Fed panics and raises rates by 100bps in two meetings.**
- **Months 5-9: The 10Y-2Y curve inverts. Retailers report a "bullwhip effect" where inventories pile up because consumers can only afford food and fuel.**
- **Months 10-12: Corporate earnings collapse 15%. Real GDP contracts by -0.5%, signaling a "Hard Landing" by early 2023.**

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5. Investment Recommendation

- **Equities: Underweight. Specifically, we recommend exiting non-profitable Technology stocks, which are the most sensitive to rising discount rates (B2 Cell).**
- **Commodities: Overweight. Energy and Agriculture act as the only viable hedge in a "Cost-Push" inflationary environment.**